

Time Series Overview

- **Definition:** A **time series** is an ordered sequence of values recorded at equally spaced intervals over time.
 - ✓ Example: Daily temperature, yearly stock prices, monthly rainfall.
- **Univariate Time Series:** Only **one value per time step**.
 - ✓ Example: Temperature at noon every day.
- **Multivariate Time Series:** **Multiple values per time step**, often related to each other.
 - ✓ Example: Temperature + humidity each day.
- **Real-world examples:**
 - ✓ Stock prices, weather forecasts, Moore's Law (transistor growth), population trends.

➤ Examples & Insights

1. **Moore's Law** → Shows exponential growth of transistors per chip over years.
2. **Spurious Correlations** → Two time series may appear related but have no real connection (e.g., arcade revenue vs. CS doctorates).
3. **Multivariate Usefulness** → Combining related variables can reveal stronger insights.
 1. **Births vs. Deaths in Japan** → Shows population decline trend.
 2. **Global Temperature vs. CO₂ Levels** → Shows strong correlation.
4. **Movement Example (Car Path):**
 1. Can be plotted as **location coordinates over time** (latitude, longitude).
 2. Acceleration changes spacing between time steps.
 3. Individual coordinates can also be tracked as univariate series.

What's Next in the Course

- **Week 1** → Understanding & preparing time series data, splitting into **train / validation / test** sets.
 - **Week 2** → Forecasting with **dense models** vs. simple numerical methods.
 - **Week 3** → **Recurrent Neural Networks (RNNs)** for forecasting (stateless vs. stateful).
 - **Week 4** → Add **convolutions** + apply on real-world data (sunspot activity).
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